

EXP 21

21. 1.Design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch.

ZERO SHOT OUTPUT

The image shows a Python script in a text editor and its execution output in the IDLE Shell 3.10.11.

Python Script:

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """
Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity.
"""

output = generator(prompt, max_length=180)

print(output[0]["generated_text"])
```

IDLE Shell 3.10.11 Output:

```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/fg.py =====
('ReadTimeoutError("HTTPConnectionPool(host='huggingface.co', port=443): Read timed out. (read timeout=10)'), ('Request ID: dd93fa95-930c-4fbc-9e2e-d9fcd8bb9c6')') thrown while requesting HEAD https://huggingface.co/gpt2/resolve/main/generation_config.json
Retrying in is (Retry 1/5).
Device set to use cpu
Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate examples to max length. Defaulting to 'longest first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you can select this strategy more precisely by providing a specific strategy to 'truncation'.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=180) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity.

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation.

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation. Provide a smart service that will help you find out how much exercise you need to do per day, and how much exercise you can do in the morning.

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation. Create a simple, easy-to-use, easy-to-use app for your Smart Fitness Watch to find your fitness goals.
```

ONE SHOT OUTPUT:

```
fg.py - C:/Users/Dhanyashree/fg.py (3.10.11)
File Edit Format Run Options Window Help
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = ""
Example

Product: Wireless Earbuds

Description:
Experience crystal-clear sound with Bluetooth 5.3 connectivity, noise canceling, and a secure fit.

Now write a similar product description for a Smart Fitness Watch.

'''

output = generator(prompt, max_length=220)

print(output[0]["generated_text"])

>>>
ITS health tracking, battery life, GPS, and connectivity.

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation.

Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation. Provide a smart service that will help you find out how much exercise you need to do per day, and how much exercise you can do in the morning.

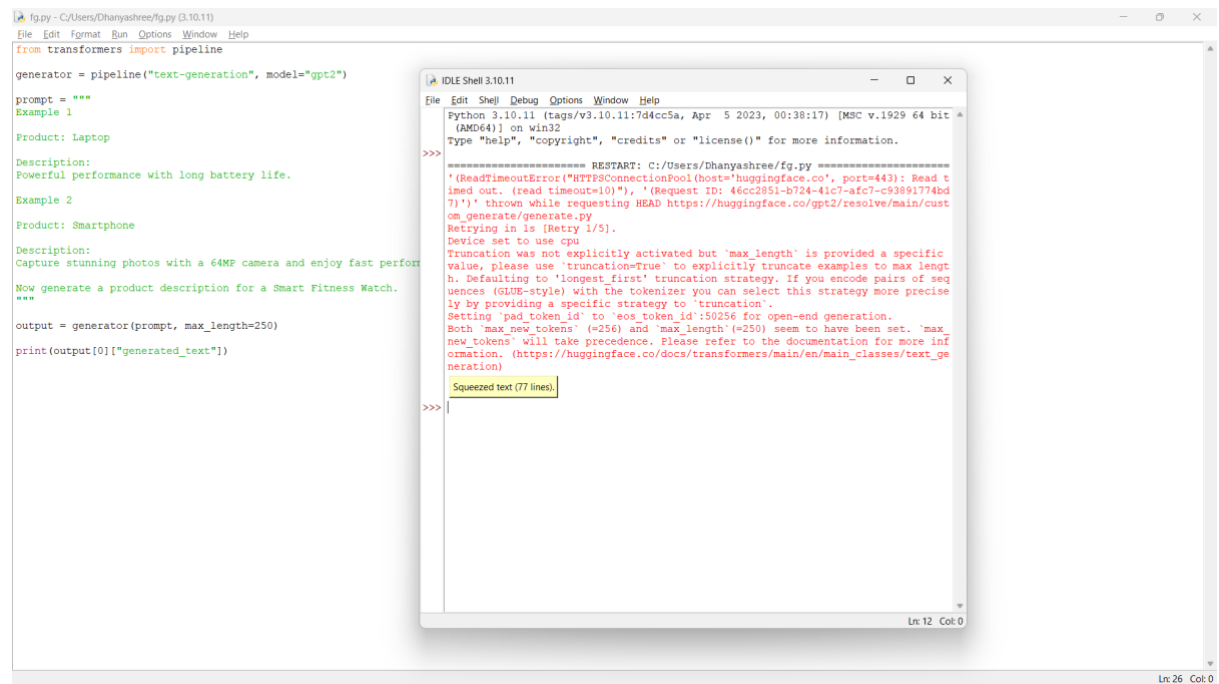
Write a professional product description for a Smart Fitness Watch highlighting its health tracking, battery life, GPS, and connectivity. Provide a professional service that will help you find the best fit for your situation. Create a simple, easy-to-use, easy-to-use app for your Smart Fitness Watch to find your fitness goals.

Write a simple, easy-to-use, easy-to-use app for your Smart Fitness Watch to find your fitness goals. Use the FitBit app to find your fitness goals, and even track your progress.

Use the FitBit app to find your fitness goals, and even track your progress. Create a simple, easy-to-use, easy

>>>
===== RESTART: C:/Users/Dhanyashree/fg.py =====
Device set to use cpu
Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate examples to max length. Defaulting to 'longest first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you can select this strategy more precisely by providing a specific strategy to 'truncation'.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=220) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Squeezed text (76 lines).
>>>
```

FEW-SHOT OUTPUT:



The image shows a Python script in a file editor and its execution in an IDE shell. The script defines a text-generation pipeline using the GPT-2 model. It includes a prompt with two examples: a laptop and a smartphone, followed by a request to generate a description for a smart fitness watch. The script then prints the generated text.

```
fg.py - C:/Users/Dhanyashree/fg.py (3.10.11)
File Edit Format Run Options Window Help
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = """
Example 1
Product: Laptop
Description:
Powerful performance with long battery life.

Example 2
Product: Smartphone
Description:
Capture stunning photos with a 64MP camera and enjoy fast performance.

Now generate a product description for a Smart Fitness Watch.
"""

output = generator(prompt, max_length=250)
print(output[0]["generated_text"])
```

The IDE shell shows the execution of the script. It displays a restart message, a warning about a timeout error, and a message indicating that the device is set to use the CPU. The output of the script is a generated description for a smart fitness watch, which is displayed in a yellow box.

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/fg.py =====
'(ReadTimeoutError("HTTPConnectionPool(host='huggingface.co', port=443): Read t
imed out. (read timeout=10)"), '(Request ID: 46cc2851-b724-41c7-afc7-c93891774bd
7)')' thrown while requesting HEAD https://huggingface.co/gpt2/resolve/main/cust
om_generate/generate.py
Retrying in 1s [Retry 1/5].
Device set to use cpu
Truncation was not explicitly activated but 'max_length' is provided a specific
value, please use 'truncation=True' to explicitly truncate examples to max lengt
h. Defaulting to 'longest_first' truncation strategy. If you encode pairs of seq
uences (GLUE-style) with the tokenizer you can select this strategy more precise
ly by providing a specific strategy to 'truncation'.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=250) seem to have been set. 'max
_new_tokens' will take precedence. Please refer to the documentation for more inf
ormation. (https://huggingface.co/docs/transformers/main/en/main_classes/text_ge
neration)

Squeezed text (77 lines):
>>>
```

Ln: 26 Col: 0